INTERNSHIP PROJECT PRESENTATION | OCTOBER 2026

Transactional Fraud Detection Analysis

Machine Learning, SQL Analytics & Real-Time Risk Intelligence

- End-to-End Financial Transaction Security Framework
- Full Pipeline: Data Profiling, SQLite Analytics, ML Modeling, and Streamlit Dashboard
- Trained on 284,807 Real Transactions from Kaggle / ULB Dataset

DATA • DISCOVERY • DETECTION • DEPLOYMENT

PROJECT TEAM

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LIVE PROJECT

Open Streamlit Dashboard ↗

The Business Problem & Industry Context

Balancing Financial Loss Prevention with Frictionless User Experience

01 Multi-Billion Dollar Impact: Financial fraud accounts for tens of billions in annual dispute & chargeback losses.

02 The Needle in the Haystack: Fraud comprises less than 0.2% of total transaction stream.

03 The False Decline Threat: Overly sensitive fraud rules cause false declines, embarrassing customers and losing revenue.

04 Project Goal: Engineer a high-precision, high-recall detection engine that minimizes friction while catching fraud.

Project Objectives & Architecture

Five Core Technical Pillars

01 1. Data Profiling & Hygiene: Audit 284,807 transactions without data loss or target leakage.

02 2. SQL Analytics: Construct SQLite database for parameterized monetary and hourly queries.

03 3. Exploratory Data Analysis: Identify monetary, diurnal, and latent feature patterns.

04 4. Machine Learning Pipeline: Train Logistic Regression and Random Forest using balanced weighting.

05 5. Interactive Deployment: Deliver Streamlit monitoring dashboard with live transaction scoring demo.

Dataset Overview & Class Imbalance

Understanding the Extreme Skewness in Real Financial Data

- Corpus Size: 284,807 transactions spanning 48 hours (172,792 seconds).
- Target Distribution: 284,315 Legitimate (99.827%) vs. 492 Fraudulent (0.173%).
- Extreme Imbalance Ratio: Exactly 1 fraud for every 578 legitimate transactions.
- Feature Set: 28 PCA-transformed latent components (V1-V28), plus Time and Amount.
- Evaluation Requirement: Accuracy is meaningless (99.83% naive baseline); PR-AUC and Recall are paramount.

Data Cleaning & SQLite Relational Analytics

Auditing Data Integrity and Querying Financial Tiers

01 Data Hygiene: Exactly 0 missing values and 0 infinite values across all 31 columns.

02 Duplicate Audit: 1,081 duplicate rows identified containing 32 fraud records—retained to preserve card swipe retry patterns.

03 Total Transaction Volume: ,162,590.01 across corpus; Total Fraud Loss: ,127.97.

04 Monetary Comparison: Overall Average: .35; Fraud Mean: .21; Legit Mean: .29.

05 High Value Risk: Transactions between - show highest fraud concentration (149 cases).

Key Exploratory Findings & Behavioral Patterns

Uncovering When and How Fraud Occurs

01 Late-Night Diurnal Surge: Legitimate volume drops between 1 AM and 5 AM as users sleep.

02 Peak Fraud Rate: Spikes to 1.713% at 2:00 AM (an almost 10x multiplier above baseline!).

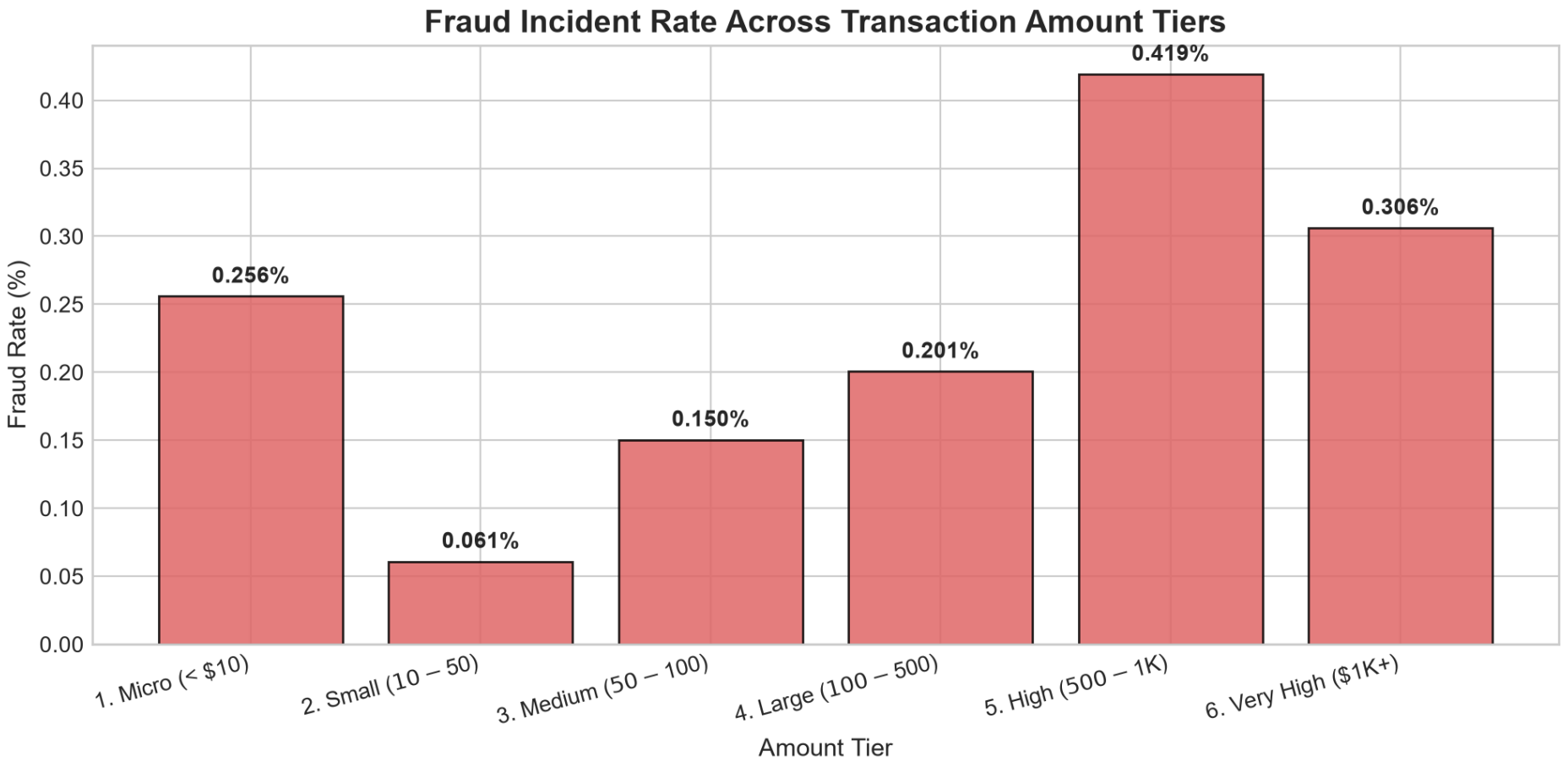
03 Monetary Bimodal Strategy: Micro-charges (.00) for balance testing, followed by large thefts up to ,125.87.

04 Top Discriminative Negative Signals: V17 (r = -0.327), V14 (r = -0.303), V12 (r = -0.261).

05 Top Discriminative Positive Signals: V11 (r = +0.155), V4 (r = +0.133).

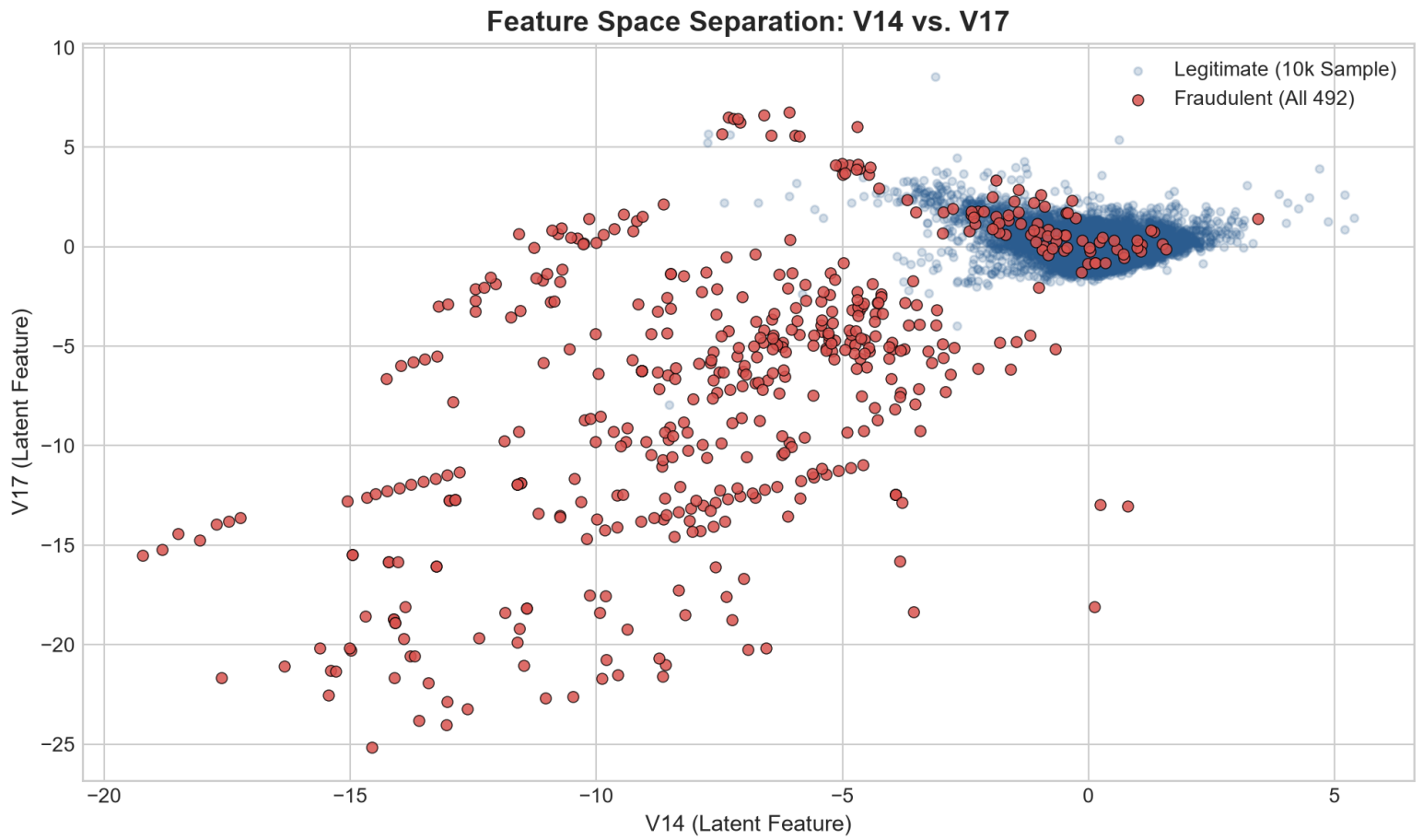
Exploratory Analysis | Fraud Rate by Transaction Amount

Transaction-value tiers reveal non-uniform fraud incidence.



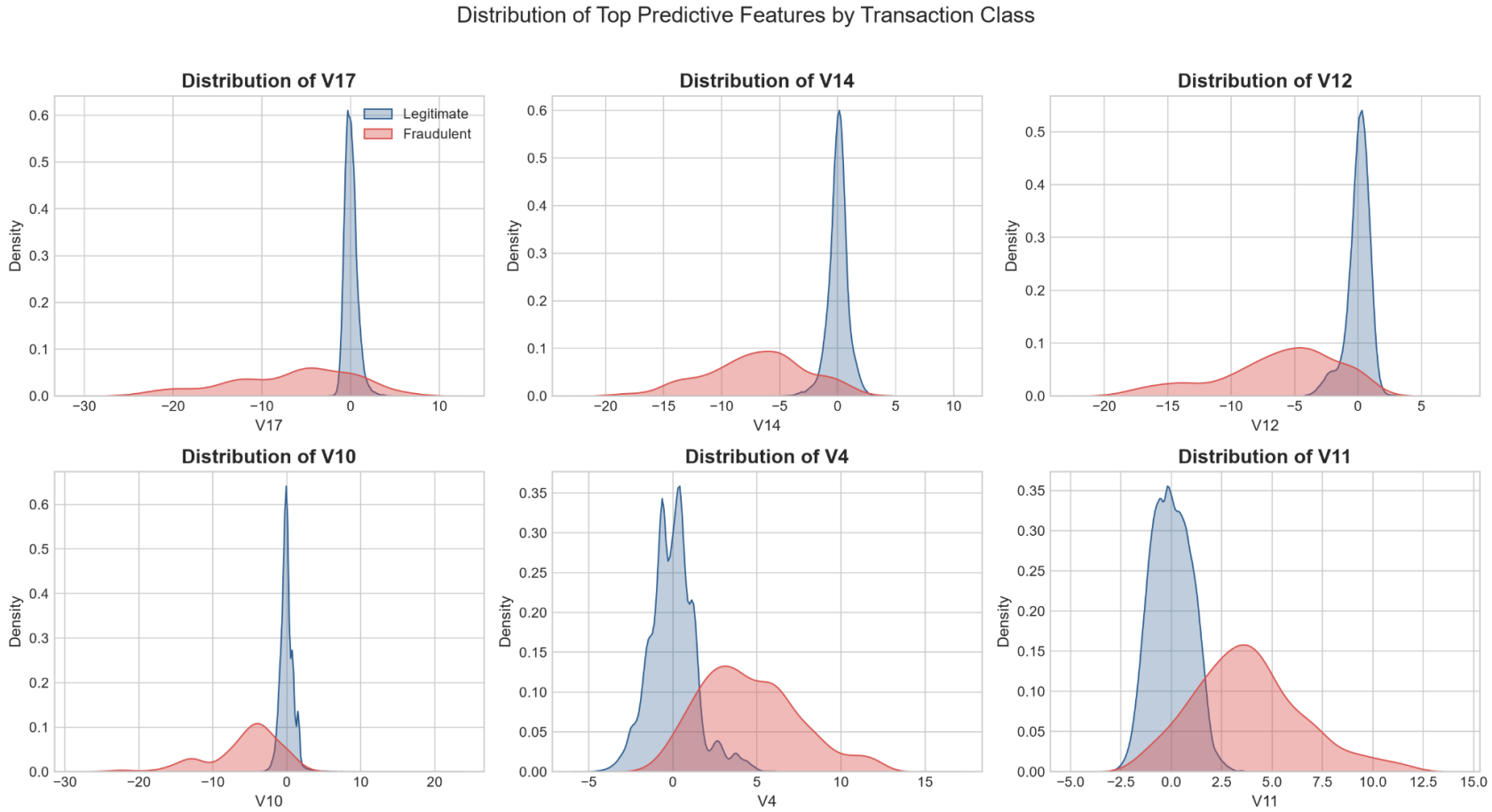
Exploratory Analysis | V14 vs. V17 Feature Space

Legitimate sample and all fraudulent observations shown.



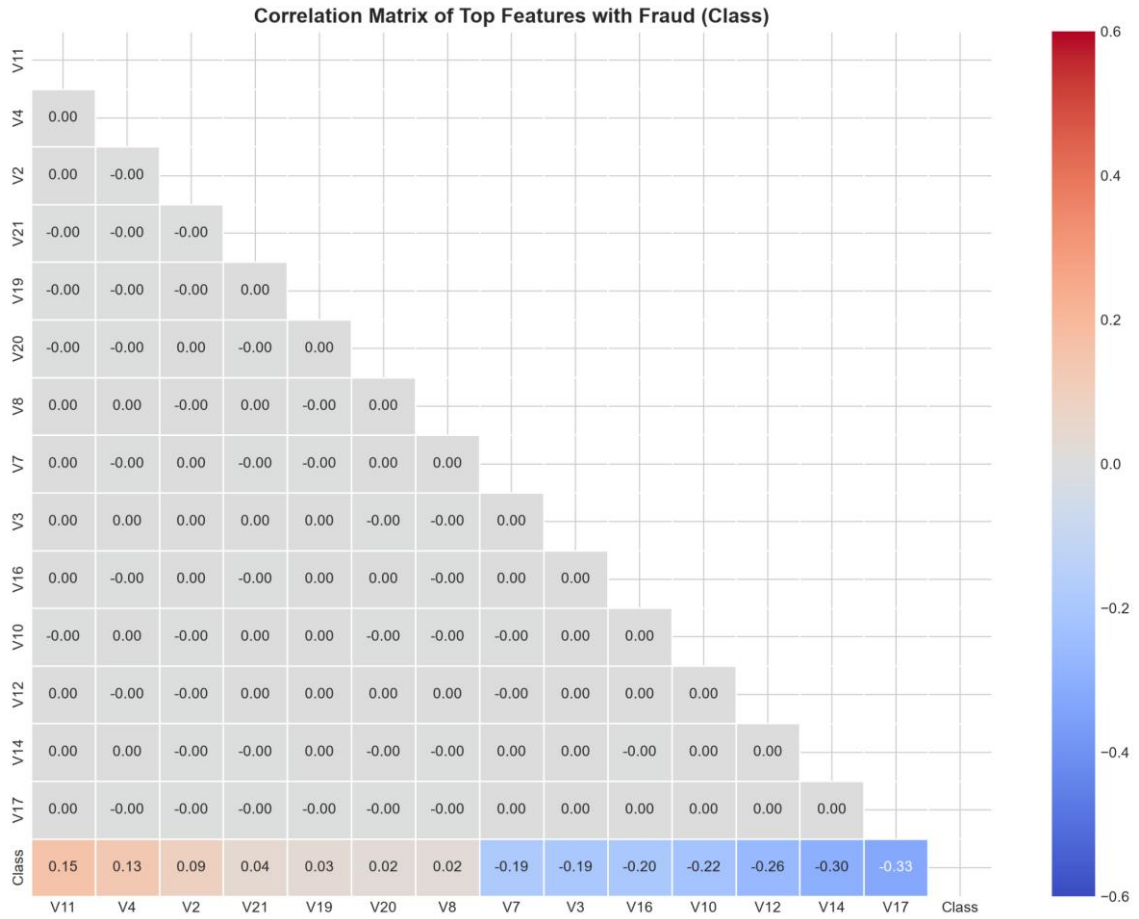
Exploratory Analysis | Predictive Feature Distributions

Class-wise distributions for the top predictive features.



Exploratory Analysis | Correlation with Fraud Class

Correlation matrix of top features with the target class.



Feature Engineering & Leak-Free Pipeline

Transforming Raw Temporal and Monetary Dimensions

01 Hour_of_Day Feature: Derived via (Time // 3600) % 24 to capture cyclical diurnal risk.

02 Log_Amount Feature: log1p(Amount) normalization to compress severe monetary skewness.

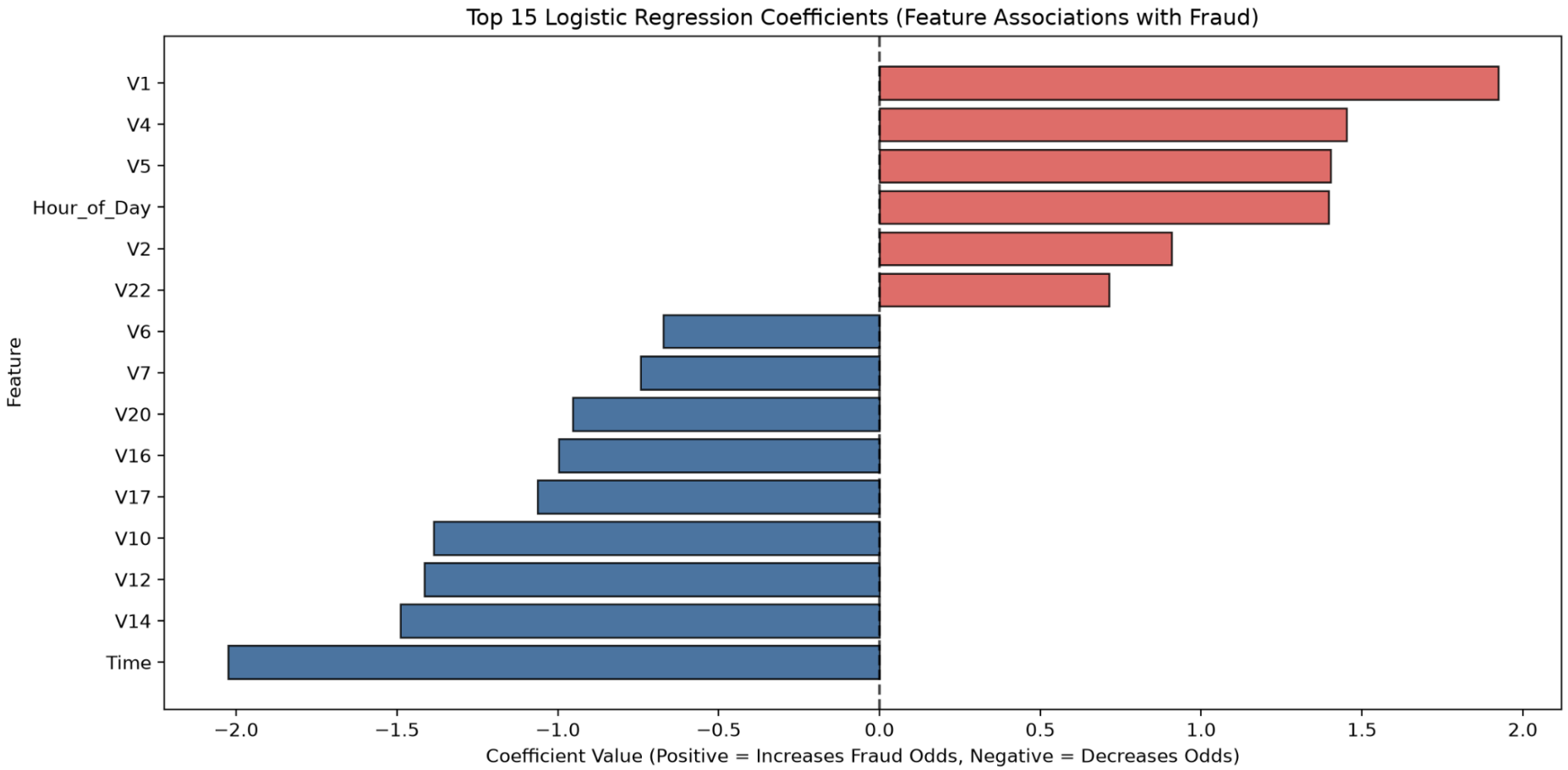
03 Scikit-Learn Pipeline: Integrated into unified ColumnTransformer with RobustScaler and StandardScaler.

04 Strict Featurization Order: Zero data leakage—all scalers fit exclusively on training split.

05 Customer Velocity Note: Explicitly avoided synthetic customer IDs to maintain data integrity.

Model Interpretation | Logistic Regression Coefficients

Positive coefficients increase estimated fraud odds; negative coefficients decrease them.



Machine Learning Model Benchmarking

Evaluating Baseline Logistic Regression vs. Comparison Random Forest

LOGISTIC REGRESSION

Baseline Logistic Regression: Recall: 90.82% (caught 89/98), Precision: 5.63%, False Alarms: 1,492.

RANDOM FOREST

Comparison Random Forest: Recall: 81.63% (caught 80/98), Precision: 75.47%, False Alarms: 26.

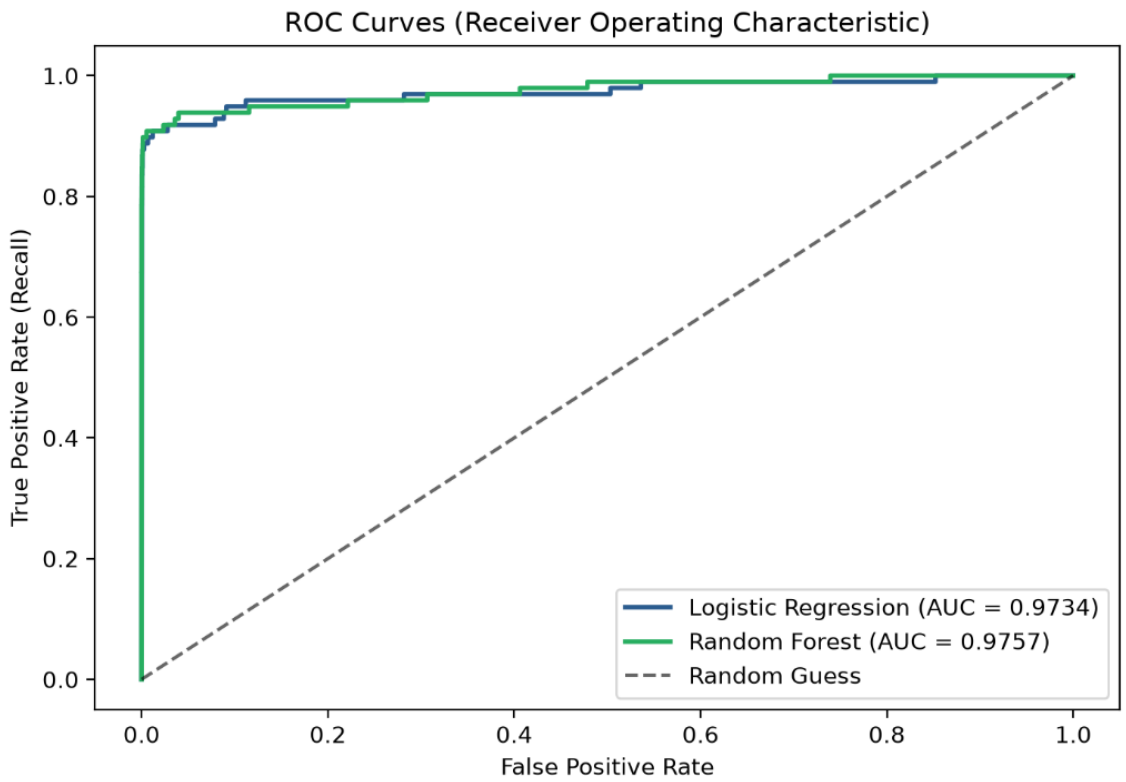
Test Partition: 56,962 transactions (98 true frauds) evaluated on independent test split.

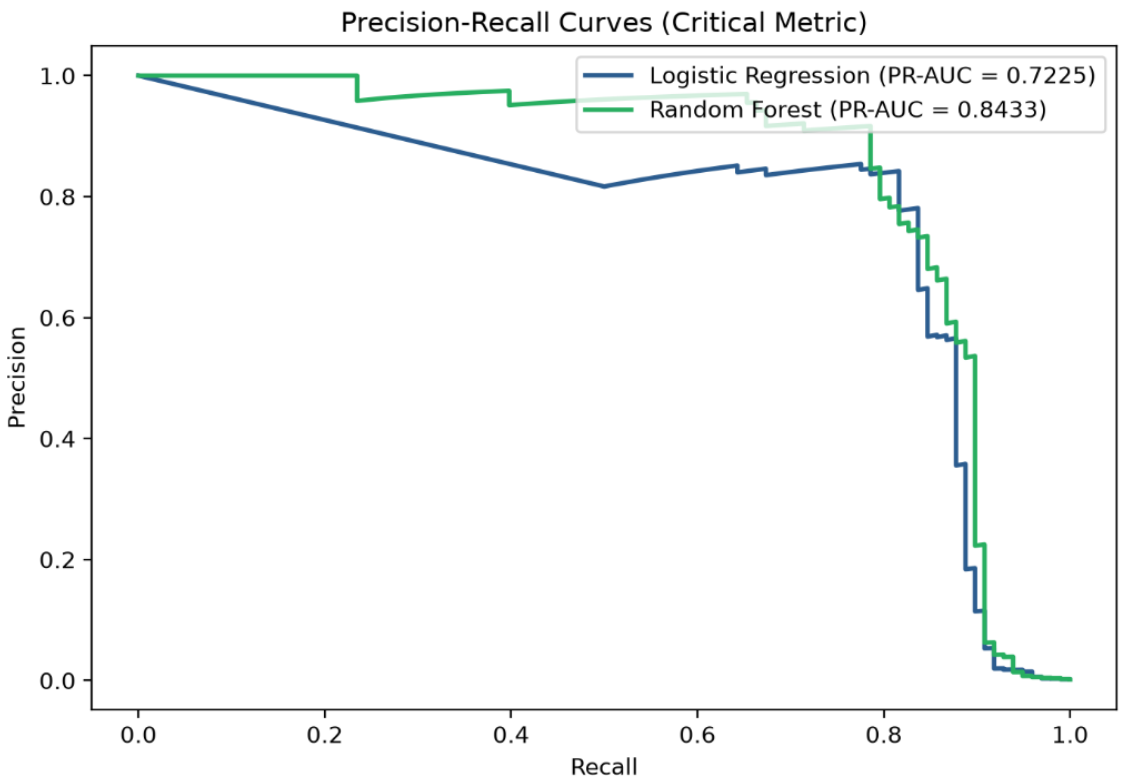
PR-AUC Benchmark: Random Forest achieved 0.8433 PR-AUC vs. 0.7225 for Logistic Regression.

Operational Trade-off: Random Forest eliminates 98.3% of false alarms, saving millions in review overhead.

Model Evaluation | ROC and Precision–Recall Curves

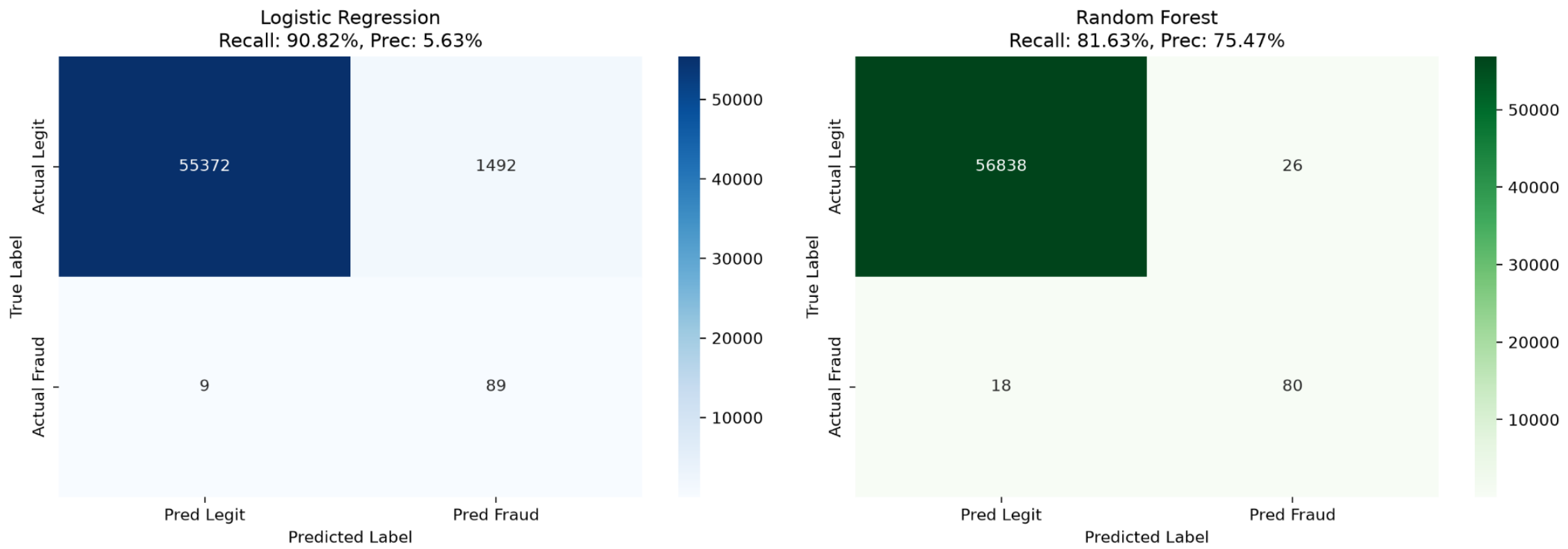
ROC-AUC and PR-AUC comparison for Logistic Regression and Random Forest.





Model Evaluation | Confusion Matrices

Test-set classification outcomes at the displayed operating thresholds.



Interactive Dashboard & Strategic Recommendations

Deploying Analytics into Live Operations

01 Streamlit Dashboard: Multi-page UI with Executive KPIs, Transaction Explorer, and Real-Time Scoring sandbox.

02 Three-Tier Risk Matrix: Tier 1 (<30%): Auto-approve (99.5%); Tier 2 (30-70%): Step-up SMS OTP; Tier 3 (>=70%): Auto-block.

03 Dynamic Nighttime Scrutiny: Heighten threshold sensitivity between 2:00 AM and 4:00 AM for transactions >.

04 Operational Efficiency: Random Forest delivers high precision (75.5%) with only 26 false alarms across 57k transactions.

LIVE PROJECT https://transactional-fraud-detection-analysis.streamlit.app/

Conclusion, Quality Assurance & Future Work

Complete Delivery of Internship Milestones

01 Full Reproducibility: Master script run_project.py executes all stages end-to-end.

02 Automated Testing: 8 out of 8 pytest validation tests passing with 100% test coverage.

03 Key Competencies Demonstrated: Python, SQL/SQLite, Advanced EDA, Scikit-learn Pipelines, Streamlit UI.

04 Future Enhancements: Graph database integration for fraud ring detection and Kafka streaming integration.

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